Network Validation and Execution Hands-on Assignment

Look up the regulatory graph of the network controlling Th lymphocyte differentiation published in Remy et al., 2006 (Figure 1). Build the corresponding Petri net model using Petrifier and compute the steady states.

1) Are there ambiguities in the regulatory graph presented in Figure 1?
2) Compare and comment the steady states you found using Petrifier with the ones published by Remy et al., 2006 (Table 1). Do they differ? Why?
3) Is it necessary to include degradation in your model to produce biologically correct predictions?
4) Try to extend the model using the latest information about the Th lymphocyte differentiation available on recent literature.

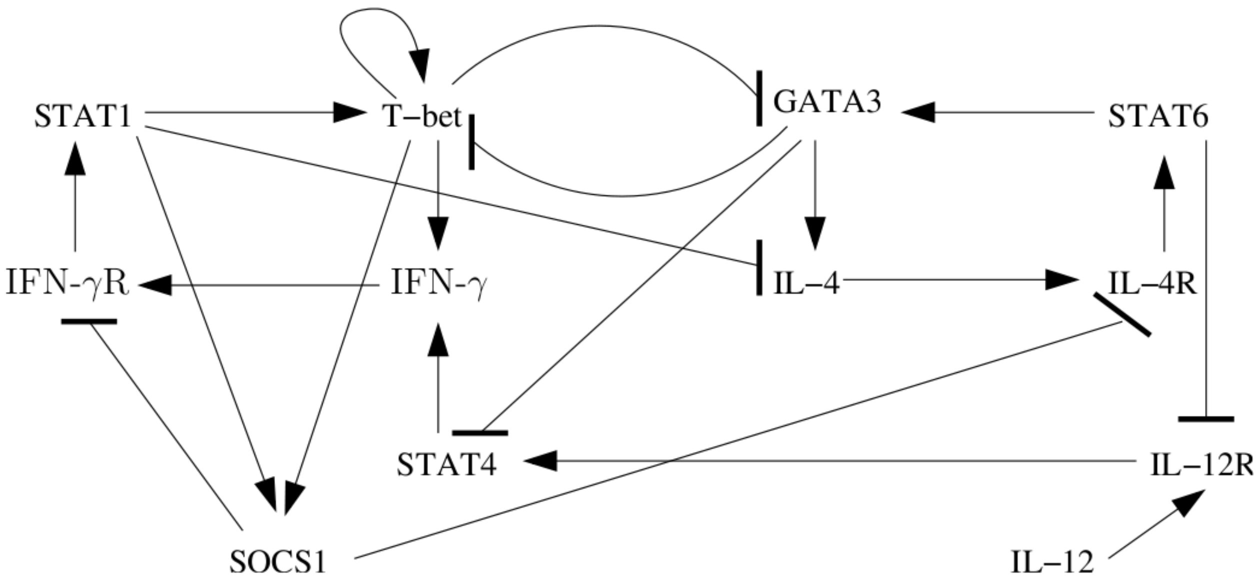

Figure 1: Th lymphocyte differentiotion gene regulatory network.

E. Remy, P. Ruet, L. Mendoza, D. Thieffry, and C. Chaouiya (2006). *From Logical Regulatory Graphs to Standard Petri Nets: Dynamical Roles and Functionality of Feedback Circuits.* In Transactions on Computation Systems Biology VII (TCSB) **4230:** 55-72 http://dx.doi.org/10.1007/11905455_3

Genes		Stable states		
		S_1	S_2	S_3
IFN-γ	(1)	0	0	1
IL-4	(2)	0	1	0
IL-12	(3)	0	0	0
IFN-γR	(4)	0	0	0
IL-4R	(5)	0	1	0
IL-12R	(6)	0	0	0
STAT1	(7)	0	0	0
STAT6	(8)	0	1	0
STAT4	(9)	0	0	0
SOCS1	(10)	0	0	1
T-bet	(11)	0	0	1
GATA-3	(12)	0	1	0

Table 1: Network steady states according to Remy et al., 2006.